

Parth Gupta

p5gupta@ucsd.edu | [Personal Website](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of California San Diego
B.Sc in Mathematics - Computer Science

San Diego, CA
Sep. 2021 – Mar 2025

EXPERIENCE

Compliance Analyst

April 2026 – Present

Bank of India

New York, NY

- Analyzed 750+ AML/EDD transactions, completing 15–20 alerts and cases daily and resolving 25% of assigned monthly workload using AML/EDD software and Accuity World Compliance.
- Prepared and transmitted ~ 30 SWIFT messages to facilitate secure cross-border financial communications and compliance tracking.
- Identified and escalated 2 potentially suspicious transactions, documenting AML/TF red flags to support SAR review and mitigate financial-crime risk.
- Performed due diligence on 750+ company profiles by analyzing dashboard, company's website, sales data to validate transaction legitimacy.

Field Researcher

December 2025 – Present

University of Michigan

New York, NY

- Conducted door-to-door housing surveys and entered structured data, achieving ~24% response rate (70+ units, 5+ pitches/unit) to build accurate datasets.
- Completed probing training while maintaining excellent data entry accuracy - personal avg. 1.33%.
- Researched building details (units, floors, authorities) to optimize interview productivity ~2.5x.

Software Engineer Intern

Jan 2024 – May 2024

More Seconds | *MongoDB, React, TypeScript, Node.js, Express.js, Docker, OpenAI, Python*

Remote, USA

- Dockerized development workflows to standardize environments and speed up deployment.
- Analyzed dashboard data to deploy AI agents, increasing task allocation accuracy by 30%.
- Developed and validated APIs supporting reliable data pipelines, ensuring accuracy and reproducibility.
- Built full-stack features (follower views, billing toggles) improving project workflows by 60%.
- Used Git and Agile methods to deliver iterative software updates on schedule.

Full Stack Intern

Aug. 2022 – Sep 2022

ThatNeedle.com | *JavaScript, Flask, HTML, CSS*

Remote, India

- Engineered backend pipelines to process and serve multi-year stock trend data and enabling dynamic sector-wise comparisons via **Chart.js** visualizations, driving a 30% increase in platform engagement.
- Built an interactive geo heat map with **Leaflet** and **D3.js** to analyze global population disparities, allowing stakeholders to identify high-risk regions for targeted resource allocation.
- Optimized visual layout and color schemes to improve readability and user engagement.

PROJECTS

Credit Card Fraud Detection | *Python, NumPy, Pandas, SciKit Learn*

- Processed 6M+ transactional records; applied SMOTE for imbalance, engineered features with MinMaxScaler & LabelEncoder, and built fraud detection models using Logistic Regression & Random Forest.
- Upgraded model from Logistic Regression to Random Forest, improving accuracy from 92.4% to 99.9% and F1-score from 6.1 to 88.7.

TECHNICAL SKILLS

Data Analysis & Programming: Python (pandas, NumPy, Matplotlib, Scikit-learn), R, SQL (queries, JOINS, CTEs, aggregates), JavaScript

Visualization & BI Tools: Tableau, Excel (advanced formulas, pivot tables, data transformation)

Databases & Data Handling: MySQL, MongoDB, NoSQL, REST APIs, data pipelines, data cleaning/validation, feature engineering

Statistics & ML Basics: Probability/Statistics, Linear Algebra, Data Mining

Cloud & Tools: AWS, Docker, Git, Google Data Analytics

Relevant Coursework: Data Structures, Discrete Mathematics, Data Mining, Numerical Optimization, Linear Algebra and Acturial Mathematics